YIFEI SUN

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Senior Software Engineer · Agentic AI Infrastructure, SDK Design & Distributed Systems · 13 Years linkedin.com/in/yifei-sun-a789861b

SUMMARY

Senior platform engineer with 13 years building distributed systems, agentic AI infrastructure, and developer-facing SDKs. Recent focus: designing agent evaluation systems, MCP server architecture for tool-calling agents, and SDK workflows for observing agent behavior and measuring progress. Deep expertise in Python (typing, Pydantic, asyncio), distributed systems architecture, and cloud-native infrastructure (AWS, Kubernetes). Led cross-team redesign of a high-traffic platform API serving millions of daily requests (p99 250ms→50ms, availability 98%→99.9%). Background includes 13 years in Java-based distributed systems, bringing mature systems thinking to Python-first AI infrastructure work.

LEADERSHIP HIGHLIGHTS

- Designed agent evaluation infrastructure (Python): execution harness, behavior observation, prompt/parameter optimization, multi-provider LLM support
- Led cross-team platform API redesign: p99 5× reduction, 99.9% availability, Kubernetes-orchestrated microservices on AWS
- Built MCP server architecture for tool-calling agents; designed SDK workflows for agent orchestration and runtime evidence collection
- Drove architecture standards and observability practices across 10+ teams; mentored engineers to Principal-level scope

Technologies: Languages: Python (typing, Pydantic, asyncio, pytest), Java (13 years: Concurrency, Streams, I/O), Node.js, Kotlin · **AI/Agent Infra:** MCP protocol, LLM orchestration (multi-provider), agent evaluation workflows, structured data validation · **Infrastructure:** AWS (EC2, ECS, EKS, SQS, SNS, S3, RDS, DynamoDB), Terraform, Ansible, Kubernetes, Docker · **Data & Messaging:** Kafka, Redis, PostgreSQL, Cassandra, gRPC, GraphQL (Netflix DGS) · **Practices:** SDK/API Design, Microservices, Event-Driven, CQRS, Distributed Tracing, SLO Frameworks · Datadog, Grafana, OpenTelemetry · CI/CD

INDEPENDENT PROJECT

Agent Evaluation Platform & GenAI Decision Engine (Python)

- Built "Prompt Lab" evaluation harness for measuring agent behavior: historical replay against real data, offline backtesting, prompt/parameter search (grid & Bayesian optimization), and multi-provider LLM comparison for cost-quality tradeoffs.
- Designed structured data models (Pydantic) for agent execution traces, decision events, and evaluation metrics — enabling reproducible measurement of agent improvements over time.
- Architected a hybrid rules + multi-path GenAI engine (fast vs. deep inference) with a policy overlay, producing explainable real-time decisions with full observability and quality assessment.
- Implemented configurable policy framework with DB-backed configs, environment-specific overrides, and scheduler-driven tuning loops for continuous agent improvement.

EXPERIENCE

CarGurus — Principal Software Engineer / Technical Lead Oct 2019 – Present

High-Traffic Platform API Modernization

- Led 3 principal/senior engineers and partnered with ~10 teams across 3 departments to redesign the core platform API (~8,000 rpm, millions of daily impressions), deploying as Kubernetes-orchestrated microservices on AWS.
- Introduced an orchestration service consolidating 12 downstream integrations into 5 well-defined services with per-service circuit breakers, timeout budgets, fault isolation, and idempotent retry.
- Built automated parity-testing infrastructure comparing new vs. legacy responses at 50% production traffic, catching dozens of edge-case bugs before they reached users.

  p99: 250 ms → 50 ms (5×) **Availability: 98% → 99.9%** **Deploy: 1 hr → 3 min** (20×)
- Defined architectural standards for new microservices; instrumented with structured logging, distributed tracing, and SLO-based latency budgets per downstream dependency.

Agentic AI Platform — MCP Server & SDK Design (Python)

- Designed and implemented MCP server architecture for agent tool orchestration, exposing product capabilities to LLM-powered agents via tool-calling protocol. Sole backend engineer owning the full SDK workflow.

- Built modular plugin system where each capability is a separately registered tool — enabling agent developers to compose functionality without service-side changes. Delivered working POC in first sprint; architecture absorbed 3 rounds of requirement changes without rework.
- Expanded to designing MCP APIs for multiple conversational AI agents across the org, establishing patterns for agent-runtime integration adopted as organizational standards.

GraphQL Federation — Netflix DGS Subgraph

- Designed and integrated a Netflix DGS subgraph into the federated GraphQL gateway, exposing data to frontend, backend, and AI consumers from a single graph.
- Implemented batch data loaders (N+1 prevention) and JVM warmup strategies eliminating cold-start latency spikes.

Digital Commerce Platform — Event-Driven Architecture

- Led backend design for a 0→1 commerce product on AWS using event-driven architecture (SQS/SNS, Kafka) with Terraform-managed infrastructure. Shipped under pandemic timeline pressure.
- Built infrastructure for rapid product experimentation — launch, measure, and safely retire product bets without destabilizing core systems.

Cross-Org Platform Leadership & Standards

- Established patterns for API design, caching, and distributed orchestration adopted across multiple teams; authored design docs, RFCs, and architecture decision records.
- Mentored senior engineers on system design and path to Principal-level scope. Led 10+ technical training sessions.

Applause — Senior Software Engineer / Technical Lead

Oct 2017 – Oct 2019

Authentication & Identity Platform

- Upgraded legacy authentication to a modern Keycloak-based SSO platform on AWS, unifying auth across internal systems, enterprise partners, and external community (OAuth 2.0, SAML, social login). Zero-downtime migration via dual-auth bridge pattern.
- Built custom RBAC plugins for Kong API gateway (Python); deployed service mesh (Linkerd on Kubernetes) with mTLS for service-to-service trust.

Event Sourcing & Microservice Architecture

- Drove design and prototyping of microservice architecture based on event sourcing and CQRS; defined foundational patterns for services, events, and read/write models adopted by new domains.
- Partnered with product, support, and customer-facing teams to modernize internal workflows and enterprise integrations.

Toast Inc. — Senior Software Engineer (early team, ~20 engineers)

Sep 2014 – Oct 2017

Core Platform & Infrastructure (Early-Stage)

- Among the earliest engineers during hypergrowth; designed and implemented core backend services (Order Service, Labor Service, REST APIs) that became the platform foundation for future growth.
- Built initial AWS environment (Ansible) — provisioned infrastructure, deployment pipelines, and monitoring before a dedicated DevOps team existed. Created first automated testing framework enabling safer releases.

Third-Party Integrations & Partner Platform

- Led integrations with key partners (workforce management, payment processors), ensuring reliable bidirectional sync. Partnered with PMs, sales, and support to translate customer feedback into platform capabilities.

The MathWorks Inc. — Software Engineer II

Jul 2013 – Aug 2014

Developed internal engineering tools (Java backend) for MATLAB/Simulink workflows. Improved backend automation systems for build/test reliability and developer productivity.

EDUCATION

M.S. in Electrical and Computer Engineering — Northeastern University, Boston 2010 – 2013

B.S. in Information Science — Beijing University of Aeronautics and Astronautics 2006 – 2010